

FOOBARENGINEERING

ESP32 –SD-CAN

ESP32-SD-CAN User Guide

ESP32 with SD card and DC power supply interfaced to CAN.

ESP32-SD-CAN
www.FooBarEngineering.com

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Introduction

The FooBarEngineering ESP32-SD-CAN is a development platform that can be used with the Arduino development environment.

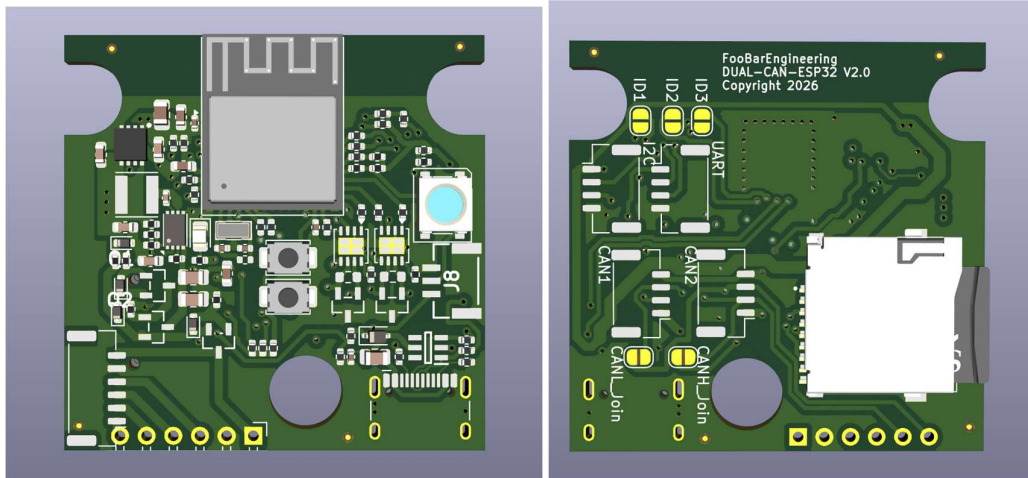
The platform has the following features.

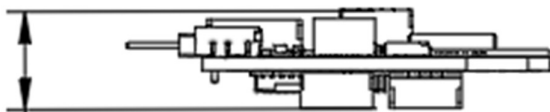
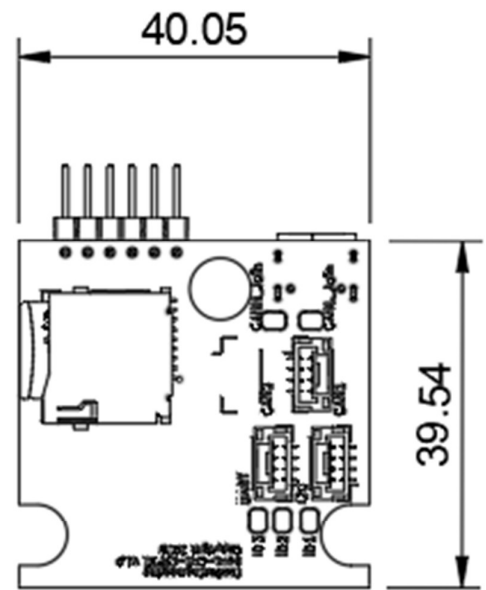
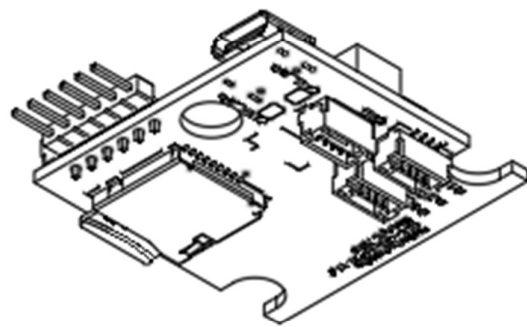
- ESP32-C6-MINI-1 module with WIFI and BLE antenna
- USBC connector
- RGB LED
- DC power supply protected for over, under and reverse voltage.
- SD card slot
- Dual CAN busses
- Ultra lightweight of 40 Grams.
- Rugged 3 point integrated anti-vibration mount design.

The ESP32-SD-CAN is intended to make prototype development using ESP32 easier to deploy as the customer can develop user code for in vehicle use.

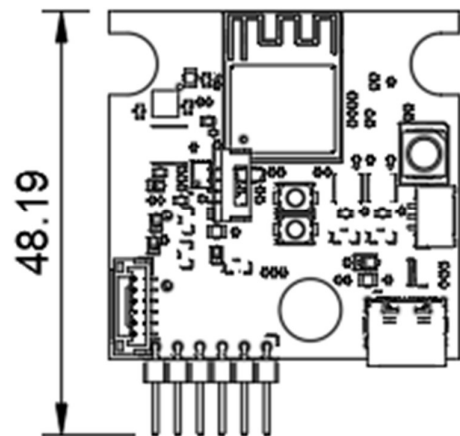
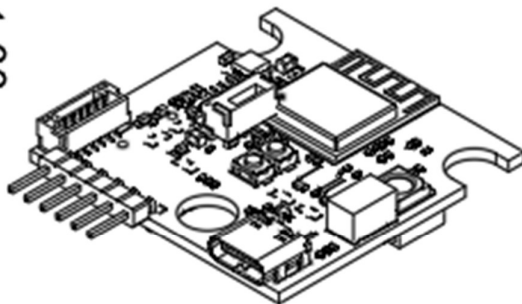
The standard enclosure is printable in small volume and uses automotive connectors to allow for a wide range of uses.

Hardware





11.08



Specifications

Electrical Data

Supply Voltage	7.9Vdc to 24Vdc
Supply Protection	Reverse and over voltage
Supply Current	300mA @ 12V
Temperature Range Operating	-10°C to +60°C
Temperature Range Storage	-20°C to +70°C

I/O Information

2x CAN	
1x USB	Connection to esp32
1x LED	RGB LED

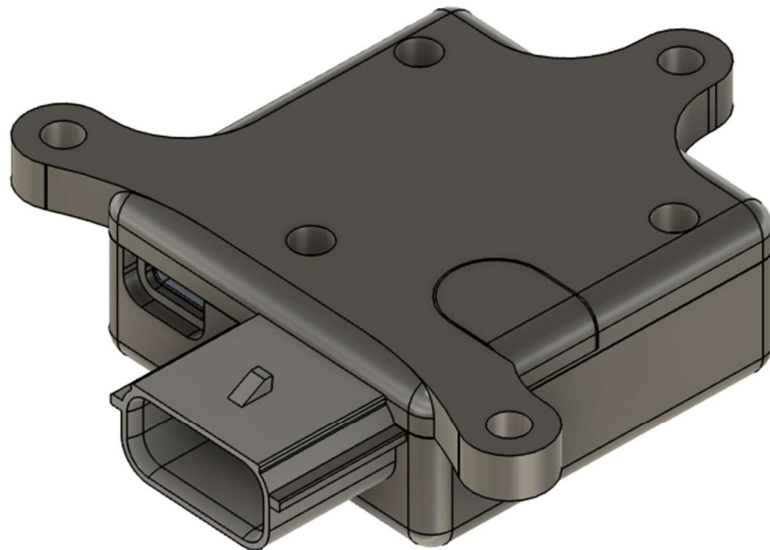
Communication Ports

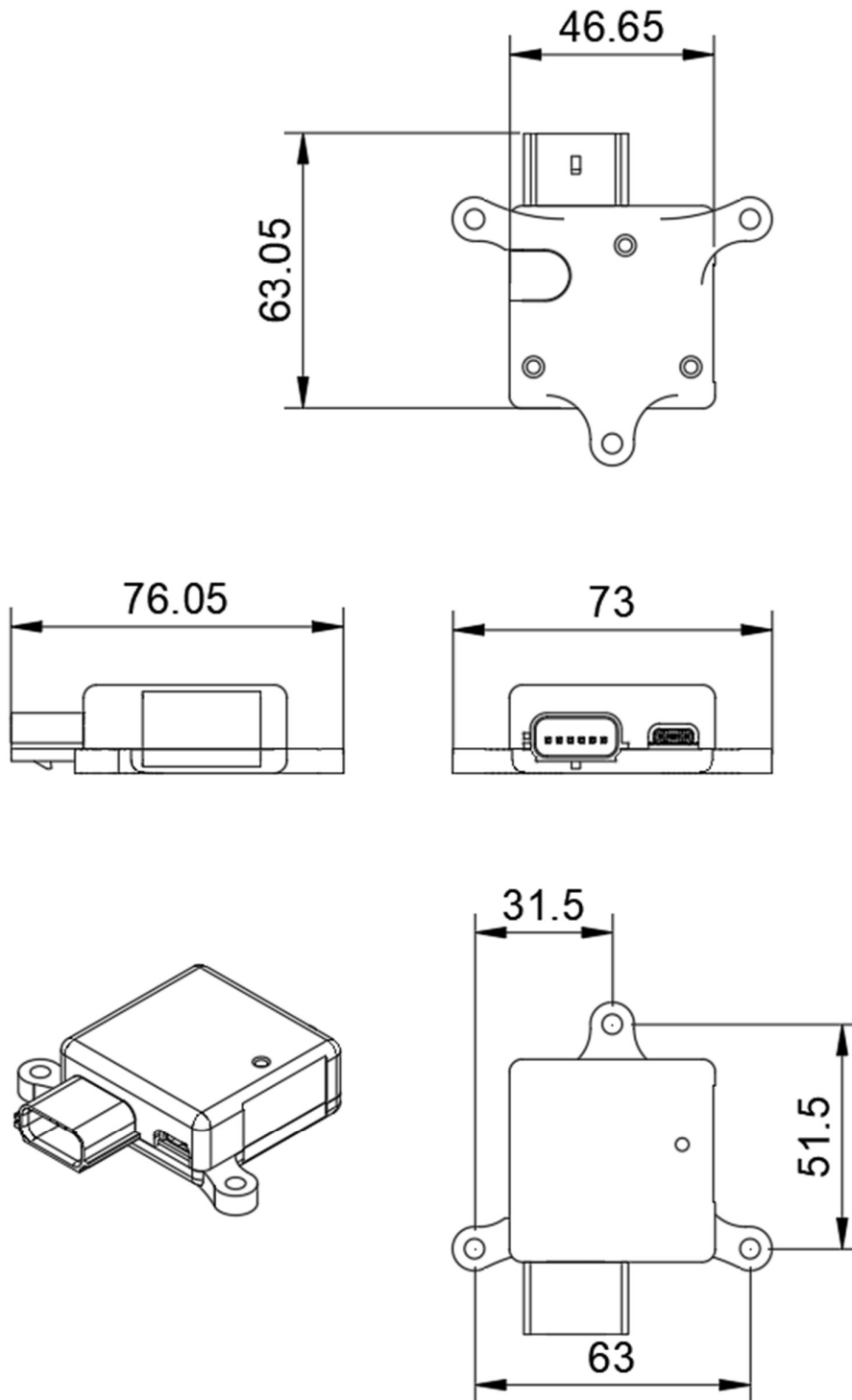
USB for Code upload from Arduino IDE

Mechanical Data

Printable enclosure

CAD model available on request for printable enclosure.





Weight 40 grams.

With connector mated unit is intended for use inside vehicles.

Wiring connectors

Mating connector Molex 31403-6100 (without CPA)

<https://www.molex.com/en-us/products/part-detail/314036100>



Signal		PIN
V Supply		1
Ground		2
CAN1 H		3
CAN1 L		4
CAN2 H		5
CAN2 L		6

Schematic

